

Air Force Life Cycle Management Center

Providing What the Warfighter Needs When They Need It

Next Generation Acquisition Model



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AFLCMC/CC
29 Aug 24**



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Reoptimizing for Great Power Competition

A Time of Consequence... A Call for Action



Gen David Allvin, CSAF

"Since the dawn of airpower, our proud heritage in history has been intertwined with key events in combat that have shaped ... a nation and in some cases, the world. Those are seminal events.

"But to me, what is more important are the times that preceded them. **These are times of consequence.** These are times when Airmen and air pioneers embodied what is running in our DNA – the **spirit of innovation, vision, and courage.**

"...What we do, the decisions we make, the actions we take, the Airmen we lead, and the way we do it, will define that next seminal event.

"We will not turn away. We will lean into it. ... The stakes are high. And the time is now."



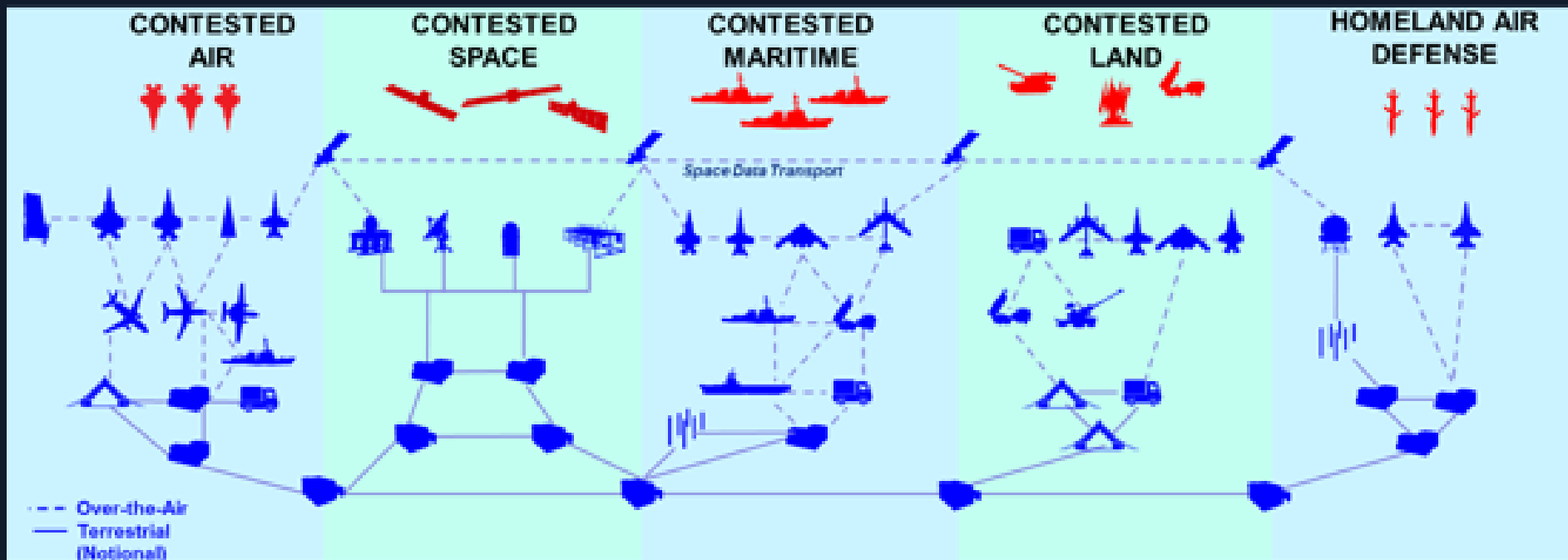
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Reoptimizing for Great Power Competition

Winning Requires System-of-Systems Approach to Warfighting

DAF Warfighting System of Systems



Reoptimizing the DAF for GPC will enable us to better design, develop, operate, and sustain a multi-domain, integrated DAF Warfighting System-of-Systems

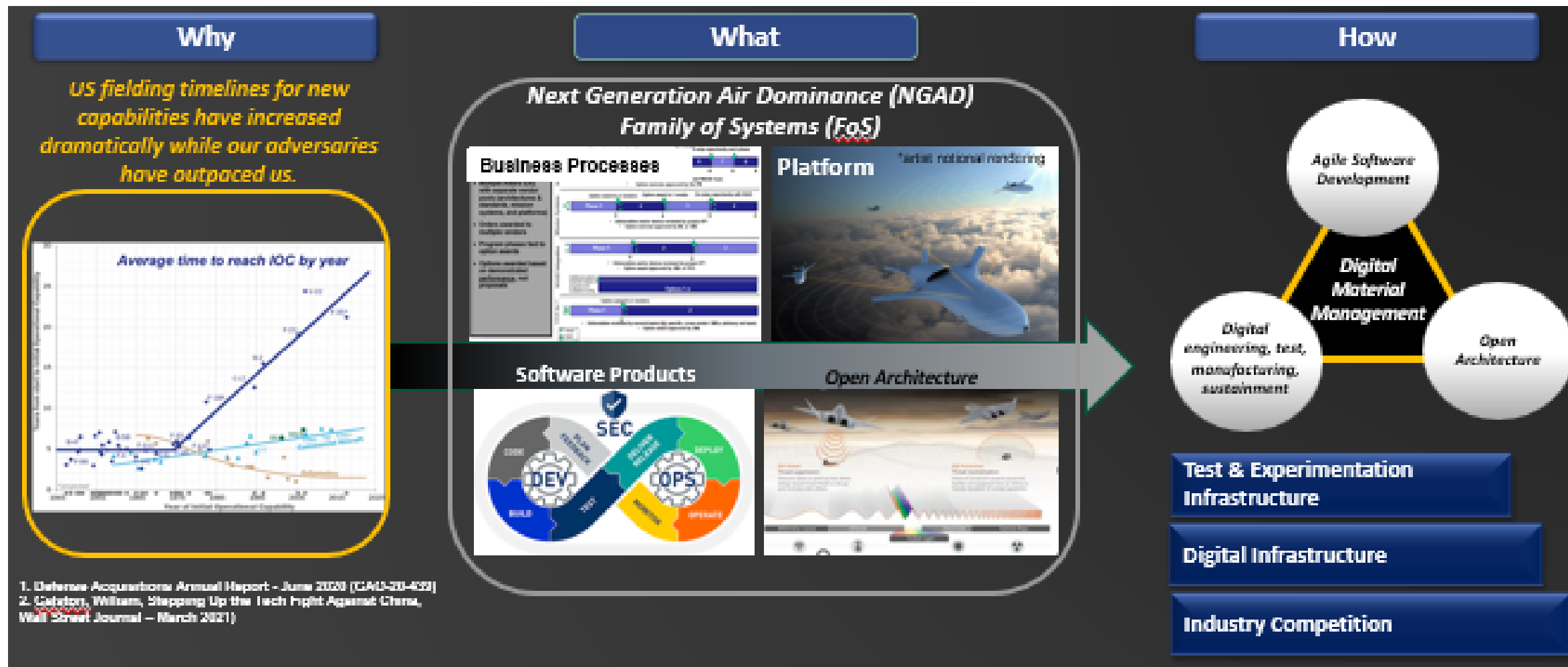


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Agile Acquisition → Next Gen Acquisition Model

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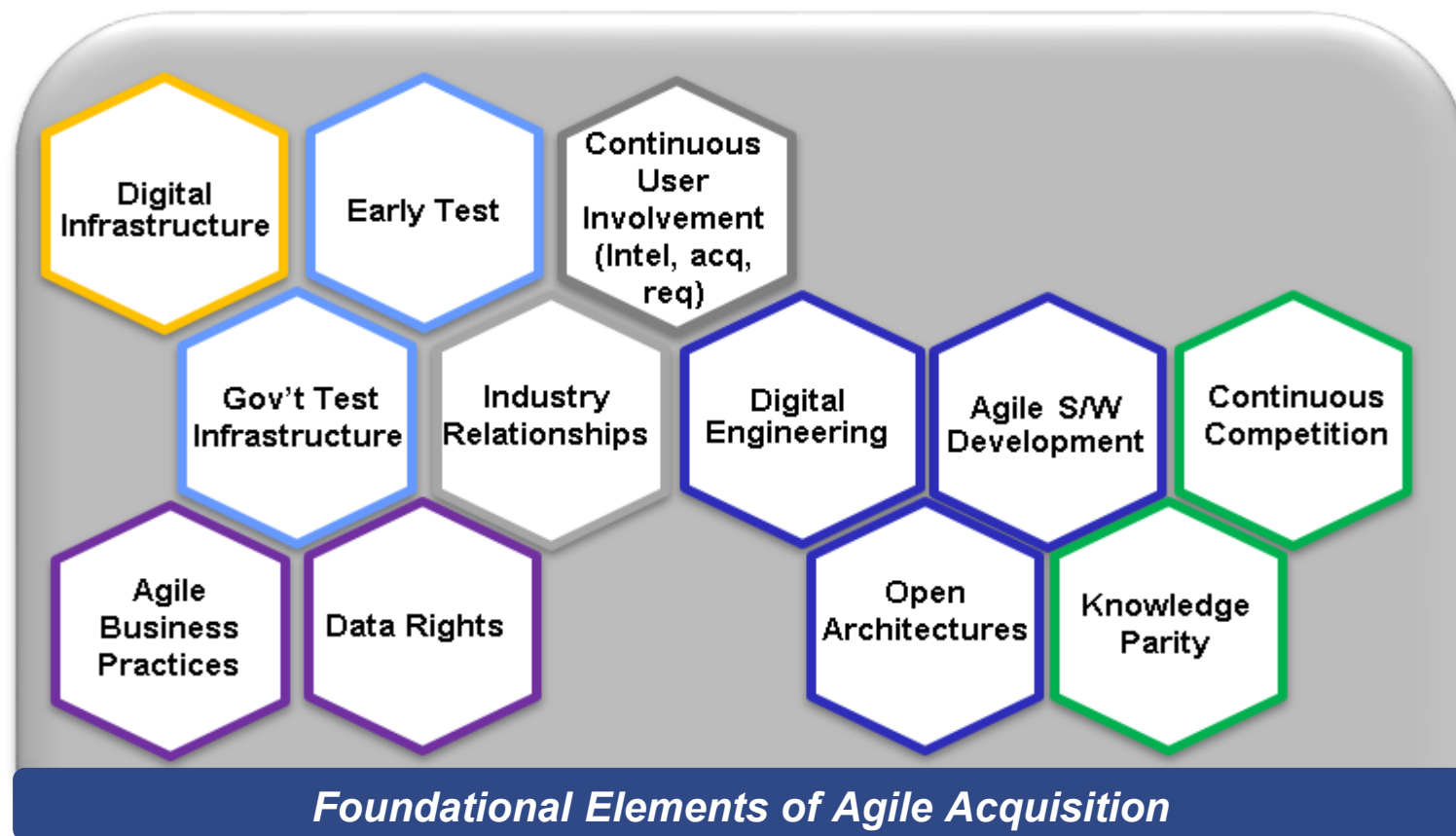
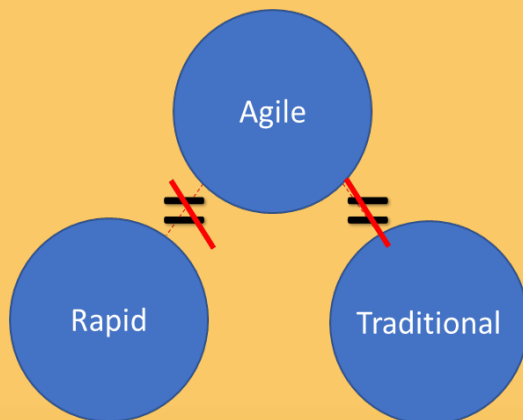


Agile Acquisition → Next Gen Acquisition Model

Agile Acquisition is:

- Acquisition approach that allows for maximum flexibility to make decisions with best info possible
- Accommodate new information or changing threat

Agile Acquisition is not:



A Powerful Paradigm Shift...Not Simply an “Acquisition Reform” Effort

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Next Gen Acquisition Model

Transparency and Communication

- Industry - Operator
- Intel - Acquisition
- S&T - Sustainment
- Test - International

Government-Owned Open Architectures and Standards

- Avoid vendor lock
- Open, common interfaces
- “Marketplace” of vendors and compliant hardware and software solutions

Key Principles

1. Early and continuous stakeholder collaboration in shared digital environment
2. Attribute-based requirements refined over time w/ data & analysis
3. Agile Acquisition Strategy: flexible, incremental approach built on open architectures
4. Mobilize industrial base, promote collaboration across marketplace

Agile Business Practices

- Contract Structure
- Supports Agile Acquisition Strategy
- On- and off-ramps
- Government Leverage

Sustained Competition

- Motivates industry to bring talent and best performance
- Promotes favorable cost schedule, and performance
- Leverages multiple vendor pools and increments
- Taps traditional and non-traditional vendors; best-of-breed solutions

The NGAD FoS Acquisition Strategy was Built upon the Foundation of this Acquisition Model



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Broader Look at Architectures & Standards

Mapping MOSA Ecosystem



Government
Owned Open
Architectures

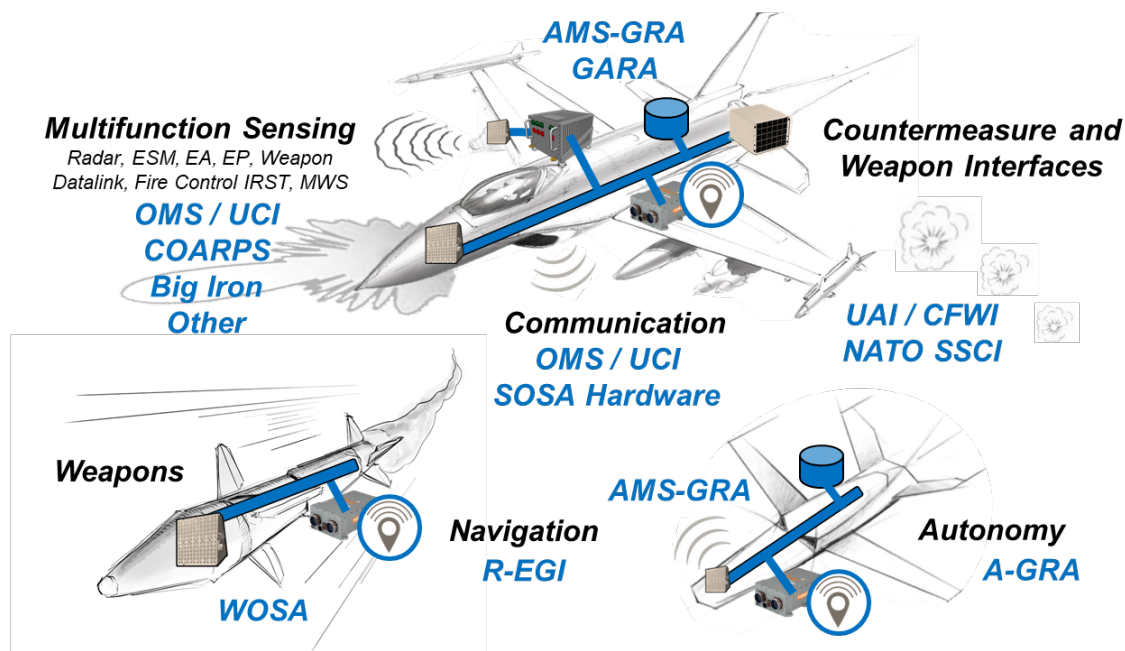
Candidate GRAs

GARA – OMS-based Gov't Ref Arch
AMS-GRA – Agile Gov't Ref Arch
A-GRA – Autonomy Gov't Ref Arch
W-GRA – Weapons Gov't Ref Arch

Candidate Open Standards

OMS – Open Mission Systems
UCI – Univ C2 Interface
UAI – Univ Armament Interface
CFWI – Common Flex Wpn Interface
WOSA – Weapons Open Sys Arch
Big Iron – EW open architecture
R-EGI – PNT open architecture
COARPS – Radar open architecture
SOSA – Sensors Open Sys Arch
SCARS – Simulator open architecture
Others – FACE, CBM+, ground sys

Kinetic Fires Grid



Objective: Identify the right candidates architectures and standards to synchronize management and realize tailored implementation informed by mission and lifecycle ROI
Business-case, Capability-case, Agility-case

How are component technologies integrated?

Black-box subsystem integrated in vendor locked OFF

Black-box subsystem integrated in Open Gov't Ref Architecture

Open subsystem integrated in vendor locked OFF

Open subsystem integrated in Open Gov't Ref Architecture

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Broader Look at Architectures & Standards

Spectrum of Open Architecture Implementation



Government
Owned Open
Architectures

F-15E Platform



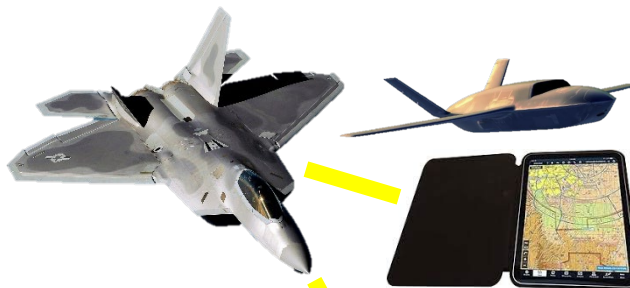
Modifications can be federated or implemented within a vendor-locked legacy system

R-EGI on F-15



Open architecture subsystems can be integrated into a vendor-locked OFP

F-22 CCA Control



A hybrid approach may use a narrow open architecture enclave within a vendor-locked system

Future Program



Future systems may integrate open subsystems into an open GRA

How are component technologies integrated?

Black-box subsystem integrated in vendor locked OFP

Black-box subsystem integrated in Open Gov't Ref Architecture

Open subsystem integrated in vendor locked OFP

Open subsystem integrated in Open Gov't Ref Architecture

Consider a range of implementation options according to lifecycle ROI analysis

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Open Architectures

Agile Mission Suite (AMS) Government Reference Architecture (GRA)

- AMS GRA is:

- Adaptable, Modular, and Expandable architecture
- Developed, Integrated, Tested, Verified CI/CD Pipeline

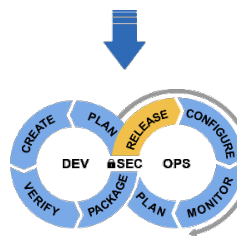
- AMS GRA enables:

- Decoupled SW from HW**
- Mission Systems isolated from safety of flight / flight critical subsystems**
- Government owned "Sensor-to-Glass"
- Not vendor lock, interfaces are open and common
- Minimizes Airworthiness impacts to deploy new capability
- "Marketplace" of Software and Hardware vendors



"NGAD FoS is Software Defined and Hardware Enabled"

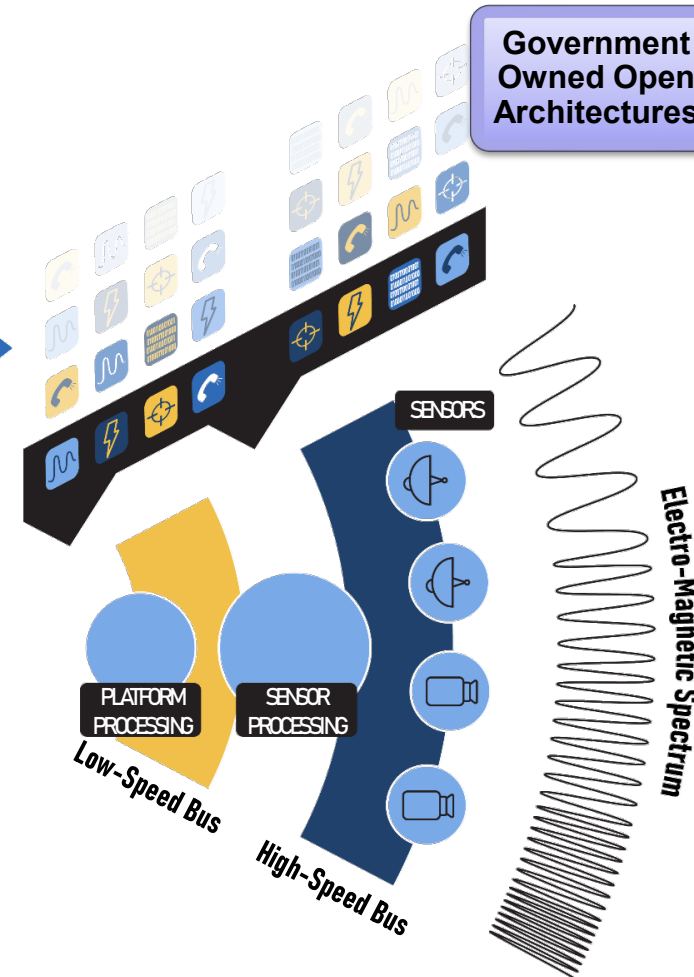
```
#include <iostream>
using namespace std;
int main() {
```



Commercial Approach



Militarized Approach



Open Architecture Ensures NGAD FoS can Respond to Dynamic Threat Environment at **Speed Of Software**

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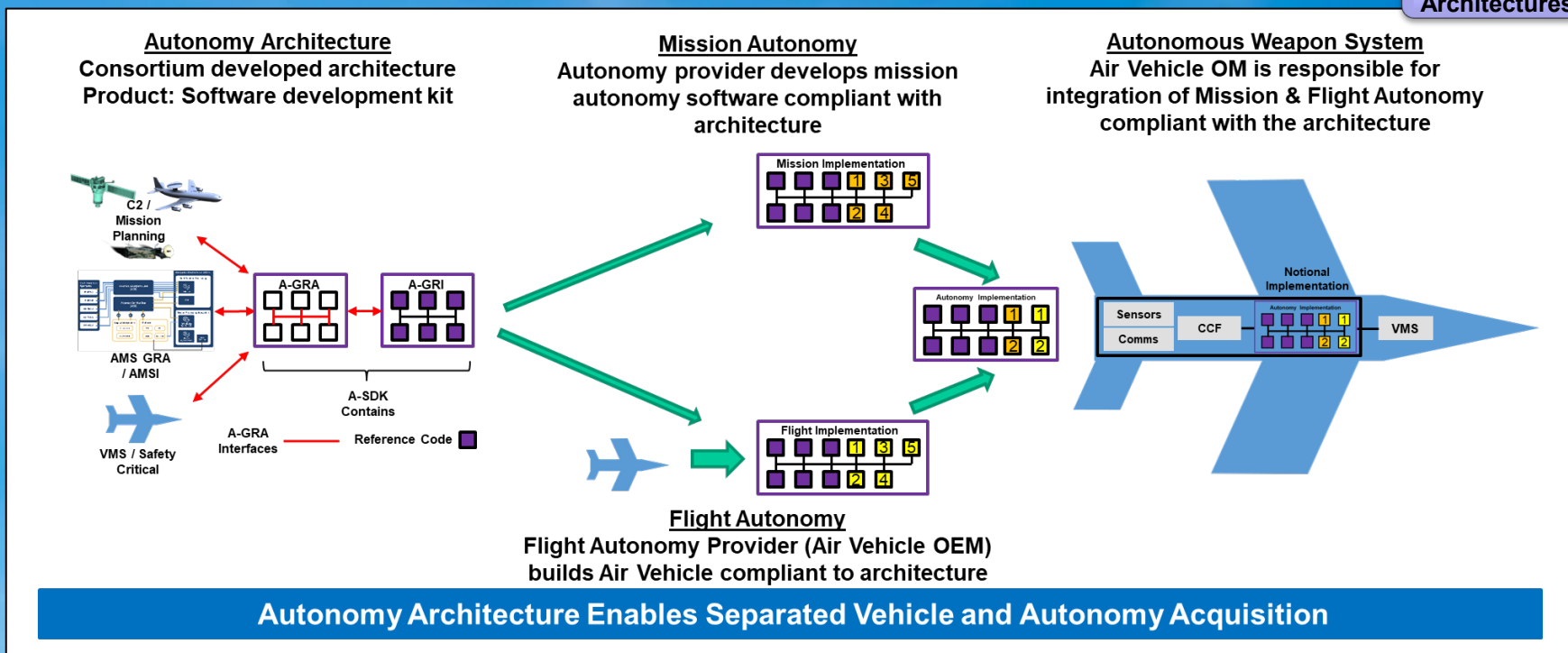


Open Architectures

Autonomy Government Reference Architecture (A-GRA)

Government Owned Open Architectures

- A-GRA is:
 - Adaptable, Modular, **Autonomy Architecture extensible to multiple platforms & missions**
 - **Affordable and scalable**
- A-GRA Enables:
 - Not vendor lock, interfaces are open and common
 - Minimizes Airworthiness impacts to deploy new capability
 - **“Marketplace” of autonomy vendors**
 - **Common user experience**
 - **Reduced manpower and logistics footprint**



A-GRA avoids vendor lock, enables rapid iteration, is extensible to other platforms, and underpins interoperability

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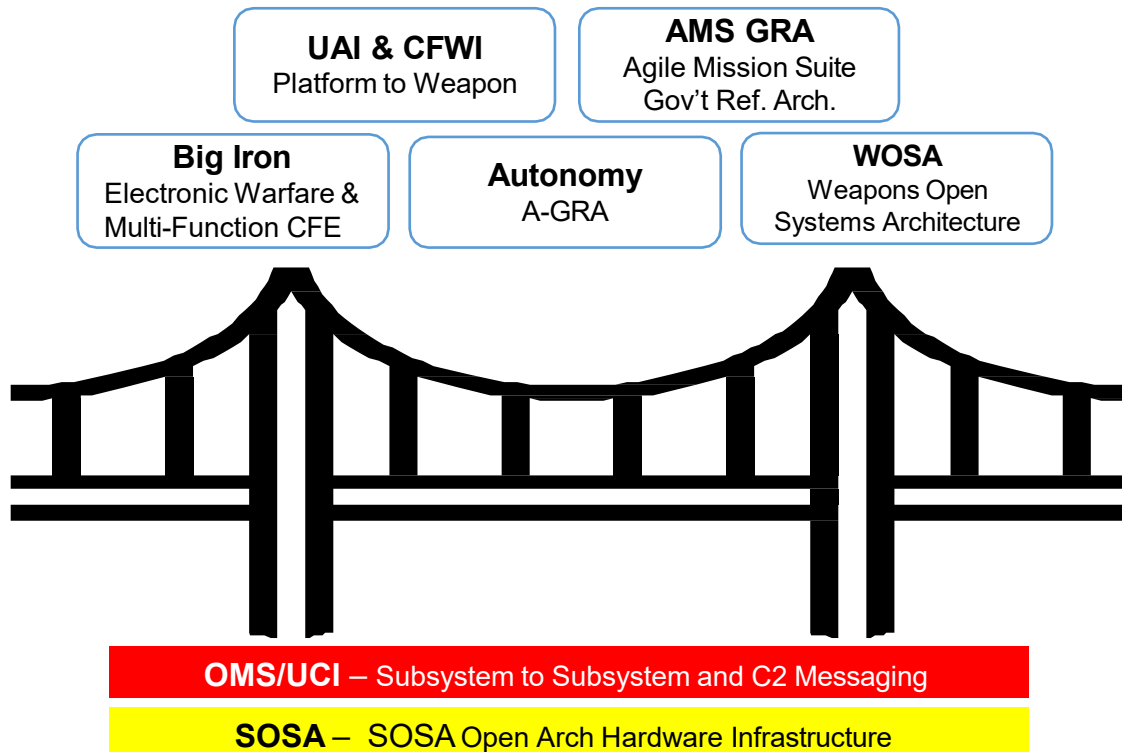


Leveraging GRAs & Standards

A Bridge from S&T to Adopting Platforms



Science & Technology



Integrated GRAs and Standards

Government
Owned Open
Architectures



**Platform & Weapon
Programs of Record**

GRAs and Standards provide the community design parameters, enabling tech transition speed and scale

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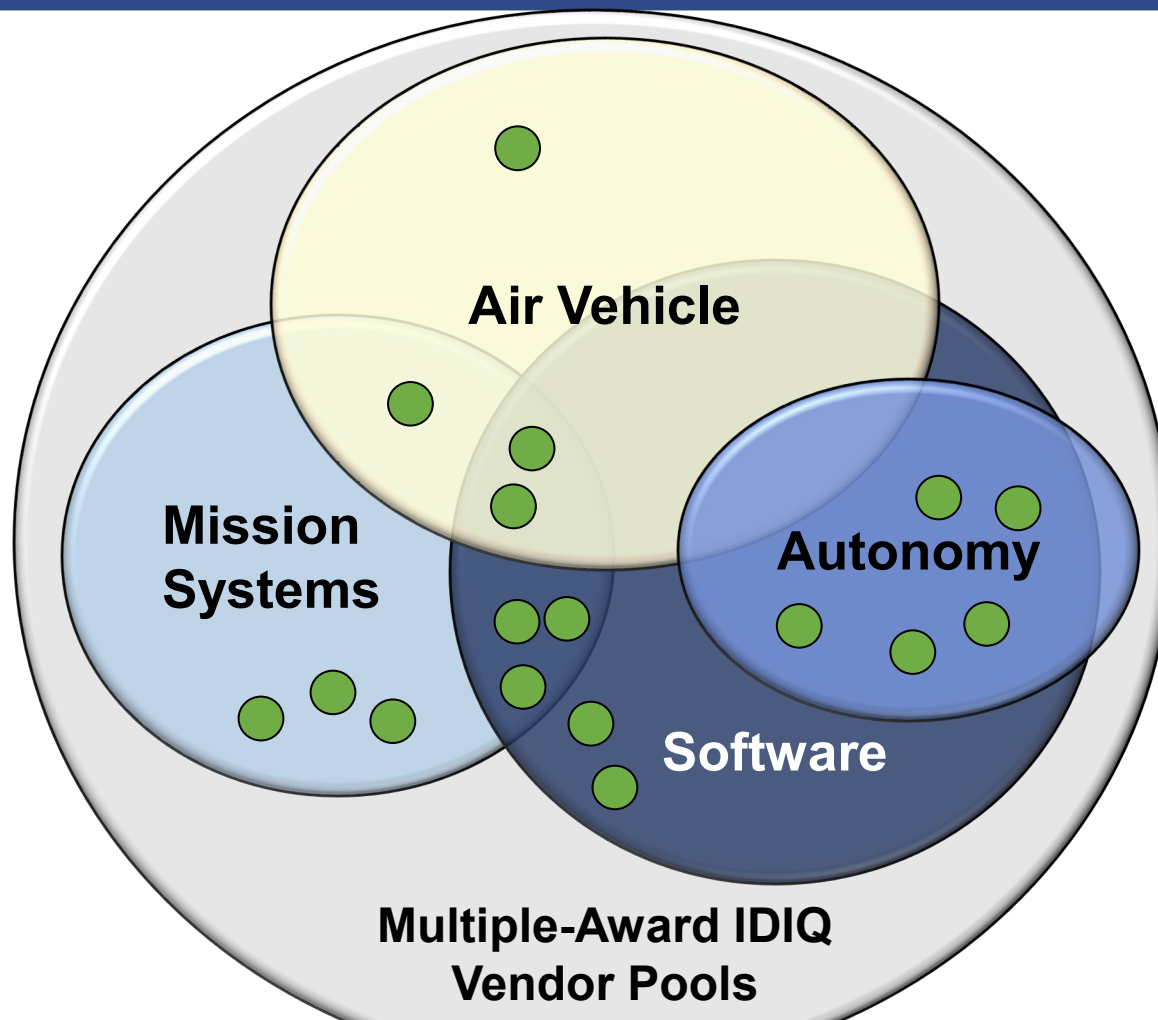


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Unlocking Sustained Competition

Sustained Competition

- Open architectures unlock competition from traditional DIB, non-traditionals and direct innovation from AFRL and tech base performers >> best of breed solutions
- Persists across multiple vendor pools, programs and increments
- Motivates industry to bring their best talent
- Promotes favorable cost, schedule, and performance



Sustained Competition

Vendor Pools

- Flexible Contractual decision space
- Marketplace for industry partners to contribute within their specialty
- Simultaneously promotes agility, inclusivity and competition

Note: Architecture adoption is critical to building a thriving marketplace >> must provide vendors in pools with opportunity to win business, not just one major contract award every 5 years.

Vendor Pools (Built on Open Architectures & Marketplace Framework) Drive Agility, Inclusivity & Sustained Competition

As of:

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Next Gen Acquisition Model: Summary

- **Enabling infrastructure:** Digital networks and engineering tools accessible across industry performers and government stakeholder sites
- **Attribute-based Requirements:** Allows you to initiate based on prioritized operational attributes refined over time based on data and analysis...instead of a delayed start after locking in fixed “waterfall” requirements...while the threat is dynamically changing
- **Stakeholder Teaming:** Tight collaboration with users, intelligence, testers, and industry experts
 - User collaboration refines product from top-level attributes
 - Stakeholder collaboration across contracts and gov’t partners to achieve outcomes, with iterative feedback along way
- **Acquisition strategy:** Continuous competition for hardware and software features built on open architecture framework and marketplace opportunities
- **Agile contracting:** Apply best contracting approach for the situation (e.g., FAR 16 Multiple-Award IDIQ with performance-based option awards, OTA, etc.)
- **Knowledge parity [difficult to scale]:** Must scale current model into a distributed architecture and systems engineering function while maintaining proximity to mission—leverage central expertise to build on agile business practice and intentionally shape open architecture adoption across programs

Next-Gen Acquisition Model offers faster business processes with more competition and government leverage...built on open architecture framework



Questions?